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PROGRAM NBODY6

*
*      N B O D Y 6++
*      *****
*
*      Regularized AC N-body code with triple & binary collisions.
*      -----
*
*      Hermite integration scheme with block-steps (V 4.0.0 April/99).
*      -----
*
*      Developed by Sverre Aarseth, IOA, Cambridge.
*      .....
*      Message Passing Version NBODY6++ for Massively Parallel Systems
*      Developed by Rainer Spurzem, ARI, Heidelberg
*
*      INCLUDE 'common6.h'
*      COMMON/STSTAT/  TINIT,NIR,NIB,NRGL,NKS
*      EXTERNAL MERGE
*
#ifdef PARALLEL
#define MPIINIT 1
#else
#ifdef ENSEMBLE
#define MPIINIT 1
#else
#define MPIINIT 0
#endif
#endif

#if MPIINIT
*      Initialize MPI
*      CALL MPI_INIT(ierr)
*      CALL MPI_COMM_GROUP(MPI_COMM_WORLD,group,ierr)
*      CALL MPI_GROUP_SIZE(group, isize, ierr)
*      CALL MPI_GROUP_RANK(group,rank,ierr)
*      PRINT*, ' This is rank=',rank, ' size=', isize, ' group=',group
#endif
*
*      Initialize the timer.
*      CALL CPUTIM(ttota)
*
*      Read start/restart indicator & CPU time.
*      IF(rank.eq.0)READ (5,*)  KSTART, TCOMP, TCRITp,
*      isernb, iserreg
*
#if MPIINIT
*      CALL MPI_BCAST(isernb,1,MPI_INTEGER,0,MPI_COMM_WORLD,ierr)
*      CALL MPI_BCAST(iserreg,1,MPI_INTEGER,0,MPI_COMM_WORLD,ierr)
*      CALL MPI_BCAST(KSTART,1,MPI_INTEGER,0,MPI_COMM_WORLD,ierr)
*      CALL MPI_BCAST(TCOMP,1,MPI_REAL,0,MPI_COMM_WORLD,ierr)
*      CALL MPI_BCAST(TCRITp,1,MPI_REAL,0,MPI_COMM_WORLD,ierr)
*
*      iserreg = max(isize,iserreg)
*      isernb = max(isize,isernb)
*      IF(rank.eq.0)PRINT*, ' iserreg, isernb=', iserreg, isernb
#endif
*
*      IF (KSTART.EQ.1) THEN
*
*      Read input parameters, perform initial setup and obtain output.

```